B. Tech. (6th Sem) Computer Science & Engineering with Specialization in AI & Machine Learning
BCSE-821 (Artificial Neural Network)

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2	0	0	**Continuous evaluation**	**40**
			End semester exam	**60**
			Total marks	**100**
			Credits	**2.0**

Course Objectives:

1. To make students familiar with basic concepts and tool used in neural networks.
2. To teach students structure of a neuron including biological and artificial.
3. To teach learning in network (Supervised and Unsupervised) for classification and regression problems.
4. To teach concepts of learning rules give design methodologies for artificial neural networks.

Unit:-1 **(7 Hours)**

Introduction: A Neural Network, Human Brain, Models of a Neuron, Neural Networks viewed as Directed Graphs, Network Architectures, Knowledge Representation, Artificial Intelligence and Neural Networks.

Learning Process: Error Correction Learning, Memory Based Learning, Hebbian Learning, Competitive, Boltzmann Learning, Credit Assignment Problem, Memory, Adaption, Statistical Nature of the Learning Process.

Unit:-2 **(7 Hours)**

Single Layer Perceptron: Adaptive Filtering Problem, Unconstrained Organization Techniques, Linear Least Square Filters, Least Mean Square Algorithm, Learning Curves, Learning Rate Annealing Techniques, Perceptron– Convergence Theorem, Relation Between Perceptron and Bayes Classifier for a Gaussian Environment.

Multilayer Perceptron: Back Propagation Algorithm XOR Problem, Heuristics, Output Representation and Decision Rule, Computer Experiment, Feature Detection.

Unit:-3 **(7 Hours)**

Back Propagation: Back Propagation and Differentiation, Hessian Matrix, Generalization, Cross Validation, Network Pruning Techniques, Virtues and Limitations of Back Propagation Learning, Accelerated Convergence, Supervised Learning Hopfield networks, Training & Examples.

Unit:-4 **(7 Hours)**

Associative memories: Linear Association, Basic Concepts of recurrent Auto associative memory: retrieval algorithm, storage algorithm; By directional associative memory, Architecture, Association encoding & decoding, Stability.

Self-Organization Maps (SOM): Two Basic Feature Mapping Models, Self-Organization Map, SOM Algorithm, Properties of Feature Map, Computer Simulations, Learning Vector Quantization, Adaptive Patter Classification.

Course Outcomes: After completion of this course, student is able to:

1. Understand the working for cognitive tasks and processing of sensorial data such as vision, image- and speech recognition, control, robotics, expert systems.
2. Create different neural networks of various architectures both feed forward and feed backward.
3. Perform the training of neural networks using various learning rules.
4. Perform the testing of neural networks and do the perform analysis of these networks for various pattern recognition applications.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. Artificial Neural Networks – B. Vegnanarayana Prentice Hall of India P Ltd 2005.
2. Fundamentals of artificial neural networks -Mohammad H. Hassoun – MIT Press ,1995.
3. Introduction to Artificial Neural systems – Jacek M. Zurada, 1994, Jaico Publ. House.

Reference Books:

1. An Introduction To Neural Networks, James A Anderson, MIT Press, 1995.
2. Neural Networks a Comprehensive Formulation, Simon S Haykin, PHI Ed.